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① Integrate the literature better

② ~~Explain more how the~~

Economic Complexity and Macroeconomic Resilience to External Financial Shocks

Economic Complexity
Research Proposal

Contents

Introduction.....	3
Data	3
Methodology	5
Expected results and applications	7
References	8

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Introduction

This research proposal explores the relationship between economic complexity and financial resilience across countries. The main aim is to assess whether greater economic complexity reduces the negative macroeconomic impact of financial stress. The analysis investigates how structural characteristics, and productive capabilities influence an economy's ability to absorb and recover from external financial disturbances.

This research topic was inspired by the findings of Emmanuel Sukadi A Sukadi in his paper titled "*Does Economic Diversification Foster Resilience to Crises? Empirical Investigation*", who investigates whether economic diversification fosters resilience to crises (Sukadi, 2022). Using local projection methods on cross-country data, that study shows that more diversified economies experience smaller output losses and faster recoveries following banking and currency crises. Building on this approach, our research aims to extend the analysis by introducing the Economic Complexity Index (ECI) as a structural measure of productive diversity. Instead of grouping countries by income level, we compare them based on their degree of economic complexity. This modification allows for a more comprehensive assessment of how knowledge-intensive and diversified production structures contribute to financial resilience, capturing also the depth of economic capabilities that influence the recovery process after financial stress.

Based on this our central hypothesis of this research is: *Economic complexity increases financial resilience*. The main problem associated with this hypothesis is that, despite the growing literature on economic complexity and growth, little empirical evidence exists on how structural complexity influences the transmission of financial shocks. Most existing studies focus on fiscal or monetary characteristics, overlooking the importance of a country's capabilities in resilience. The goal of this research is therefore to quantify how economic complexity moderates the effect of financial stress on macroeconomic performance, using a cross-country panel framework and local projection methodology to estimate the dynamic relationship between these variables.

Data

Aligning with our research question and the examined literature in connection with our topic, the chosen dependent variable for the analysis is the annual percentage growth in GDP.

The Financial Stress Index (FSI) will be used as a shock variable to capture financial tension in domestic markets. It accounts for credit spreads, the change in asset prices and in financial conditions. This is an indicator of financial instability used in other research in connection with the topic (Hites, Giovanni, Davide, Chris, & Hanbo, 2023). A non-zero index corresponds to exogenous disruption in the financial markets (f.e. decrease in asset prices). Using this variable as shock allows us to trace the effect of financial stress on the countries' macroeconomic measures. The data is constructed for 110 countries from the year 1967 to 2024, and was collected from the results of the research in the paper of Hites Ahir, Giovanni Dell'Ariccia, Davide Furceri, Chris Papageorgiou, and Hanbo Qi titled „Financial Stress and Economic Activity: Evidence from a New Worldwide Index” (Hites, Giovanni, Davide, Chris, & Hanbo, 2023).

To moderate the strength and direction of the relationship between our independent and dependent variables Economic Complexity Index (ECI) will be used in the analysis. ECI measures a country's productive capabilities indirectly by analyzing the products the country exports. Meaning Countries with higher ECI possess more diverse export baskets. The data was collected from *oec.world* and is available for 137 countries from 1998 to 2023.

To account for any endogeneity problem that may arise (such as omitted variable bias), a global exogenous instrumental variable will be used in the projection. An instrumental variable is only related to the outcome through its effect on the endogenous variable, thus we have chosen the Merrill Lynch Option Volatility Estimate (MOVE) index, which measures the volatility in US Treasury. The variable was obtained from *finance.yahoo.com* and was available from 2002 until 2025.

In the equation, inflation and unemployment rate will be included as control variables. Inflation will be used with the goal to account for macroeconomic conditions and monetary policy stance at the time of the shock, whilst the unemployment rate will control for labor market slack and cyclical conditions throughout the shocks span. Higher inflation can reflect overheating, which may amplify the effects of financial stress and weaker real returns. Controlling for this helps isolate the real impact of the shock on monetary dynamics. On the other hand, labor market conditions can influence the length and persistence of financial crises. Higher unemployment rate may cause weaker aggregate demand and slower recovery phase. Including this in the calculations ensures

that FSI shocks on output are not distorted by already existing condition in the labor market. Both figures were obtained from *data.worldbank.org* and could be found between 1960 and 2024 for 137 countries.

During the data-cleansing process we identified and removed error like inaccuracies, mostly missing information from the dataset to improve the quality and reliability of the analysis. By doing this we ended up with 36 countries indicators from 2002 to 2023. Due to the small number of observations (22) asymptotical stability should be considered when interpreting our results.

The main concerns regarding the dataset will be measurement error, where the indicators like FSI's and ECI's composition can differ across different databases and may include financial condition that are partly caused by themselves creating a so-called reverse causality. Omitted variable bias can be another issue, where there could be unobserved factors that aren't controlled for, creating bias in the estimated coefficients.

Methodology

As far as the methodology of the analysis is concerned, we have taken into consideration many possible models, keeping an eye on the purpose of the research and the desired output. Since our main goal was to reveal how economic complexity contributes to the macroeconomic resilience to financial shocks in general, our idea was to reveal the dynamic causal relationship between the countries' financial stability and economic growth and scrutinize how the effect of such external shock (from Financial Stress Index) varies with economic complexity.

The traditional approach for this is to use vector autoregression linear models and estimate the impulse responses. This methodology requires the specification of the full economic environment in order to trace the effects of external disturbance over time (Jordà, 2023). However, there is another approach, which would facilitate to estimate local projections for specific time periods (Jordà, 2005).

The key distinction between the two frameworks is that while in case of the VAR model, impulse responses are generated from a single estimated system and traced out as a continuous function over future horizons, in contrast, Oscar Jordà's local projection method estimates the response at each horizon separately by running a sequence of regressions for the endogenous variable. It has

several advantages as well, including its simplicity, robustness to misspecifications and flexibility over VAR.

Considering the above mentioned, we believe that using Local Projection would be a better method to capture the true causal relationship between our variables. As our explanatory variable (FSI) is mainly derived from domestic macroeconomic factors, there is a real possibility that due to reverse causality and confounding variables, it might be endogenous to GDP growth rates. Since endogeneity is one of the main sources of inconsistent and biased estimates due to omitted causes (Antonakis, Bendahan, Jacquart, & Lalive, 2010), we will have to deal with that arising problem.

Because of our serious concern, the use of instrumental variable and 2-Stage Least Squares method seems to be the most optimal solution to exclude the exogenous variation from the Financial Stress Index. The general framework of the latter consists of two main regressions: first we have to regress the presumably endogenous shock variable on an exogenous instrument, save the fitted estimates of the model, then we need to include those to the second stage – original – equation to express the exogenous effect on GDP growth variable. As we plan to use control variables, we need to include those as well. As previously indicated, we used the MOVE index as the instrumental variable.

$$FSI_{i,t} = \alpha + \beta_0 * MOVE_t + \gamma_1 * ECI_{i,t-1} + \gamma_2 * Inflation_{i,t} + \gamma_3 * Unemployment\ rate_{i,t} + u_t$$

Equation 1 2SLS 1st stage panel regression

$$GDPgrowth_{i,t+h} = \alpha^h + \beta_h * fitted(FSI)_{i,t} + \delta_{h,1} * ECI_{i,t-1} + \delta_{h,2} * (fitted(FSI)_{i,t} * ECI_{i,t-1}) + \gamma_{h,3} * Inflation_{i,t} + \gamma_{h,4} * Unemployment\ rate_{i,t} + u_{t+h}^h$$

Equation 2 Local projection panel regression (2SLS 2nd stage regression)

Considering the approach and goal of our research – to reveal the impact of economic complexity on the vulnerability of the observed countries to financial surprises – we would include the interaction term of the instrumented shock and ECI variables, since we believe in the dependency of those effects on one another. Although IV estimates are considered less efficient, they are typically preferred to OLS ones due to lower bias.

We can test for endogeneity by running Hausman's test, comparing the 2SLS and OLS estimations (Bettinger, 2010). It is worth noting that instrumenting only performs effectively, if the instrument is strong enough, thus it will need to be assessed as well (Keane & Neal, 2023). If we found that the MOVE index does not explain enough variation in the explanatory variable, we might need find another global exogenous instrument or use multiple ones, to prevent invalid inference. We need to check the variance inflation factor to make sure multicollinearity does not distort the results.

The projected equations and variables might change throughout the data preparation process. Impulse response analysis requires covariance-stationary input. Therefore, we will need to test for statistical stability (eg.: Augmented Dickey-Fuller test) and carry out the adequate transformations where necessary (differentiating for 1st order integrated processes). We choose our final model based on information criteria (BIC, AIC).

The analytical tool which we would use is R studio, more precisely "lpirfs" R package, which provides a comprehensive framework for building panel models and compute impulse response functions using Local Projections with shock variable identified outside of VAR (Adämmer, 2019).

As for the second part of our analysis, composing our final model, we can finally visualize the impulse response functions in such a way that we divide our responses of the countries based on ECI values in the lower (0.25) and upper (0.75) quartiles. Then we compare the shock effects in case of lower and higher values of economic complexity and draw our conclusions.

For robustness we might try different specifications:

- Use another global exogenous instrument variable.
- Use other shock variable (include dummy).
- Separate the dataset before constructing the model based on ECI, run three-four separate LP regressions and visualize IRF-s.

Expected results and applications

Regarding the expectations about the results of the analysis, we formulated those based on our studies, some superficial research and findings of previous papers on the topic. Since it has been discovered by Jose E. Gomez-Gonzalez, Jorge M. Uribe and Oscar M. Valencia that economic

complexity has a significant impact on the likelihood of that a country experiencing fiscal crisis (Gomez-Gonzalez, Uribe, & Valencia, 2023) - as higher complexity reduces fiscal vulnerability - , we believe, as our above introduced hypothesis indicates, that our observations would align with the revealed phenomenon, and the effect of the exogenous shock would be smaller (or even insignificant) for countries with the highest ECI-s compared to those at the lower end.

Our paper could contribute by not only possibly serving as a verification of the already assumed and studied relationship between financial stability, economic growth and complexity, but also connects that investigation with dynamic causal analysis using impulse response functions. Based on the findings, smaller, financially less stable economies might be more encouraged to prioritize economic complexity, not only to promote long-run growth, but also to strengthen their capacities to withstand economic and financial distress.

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